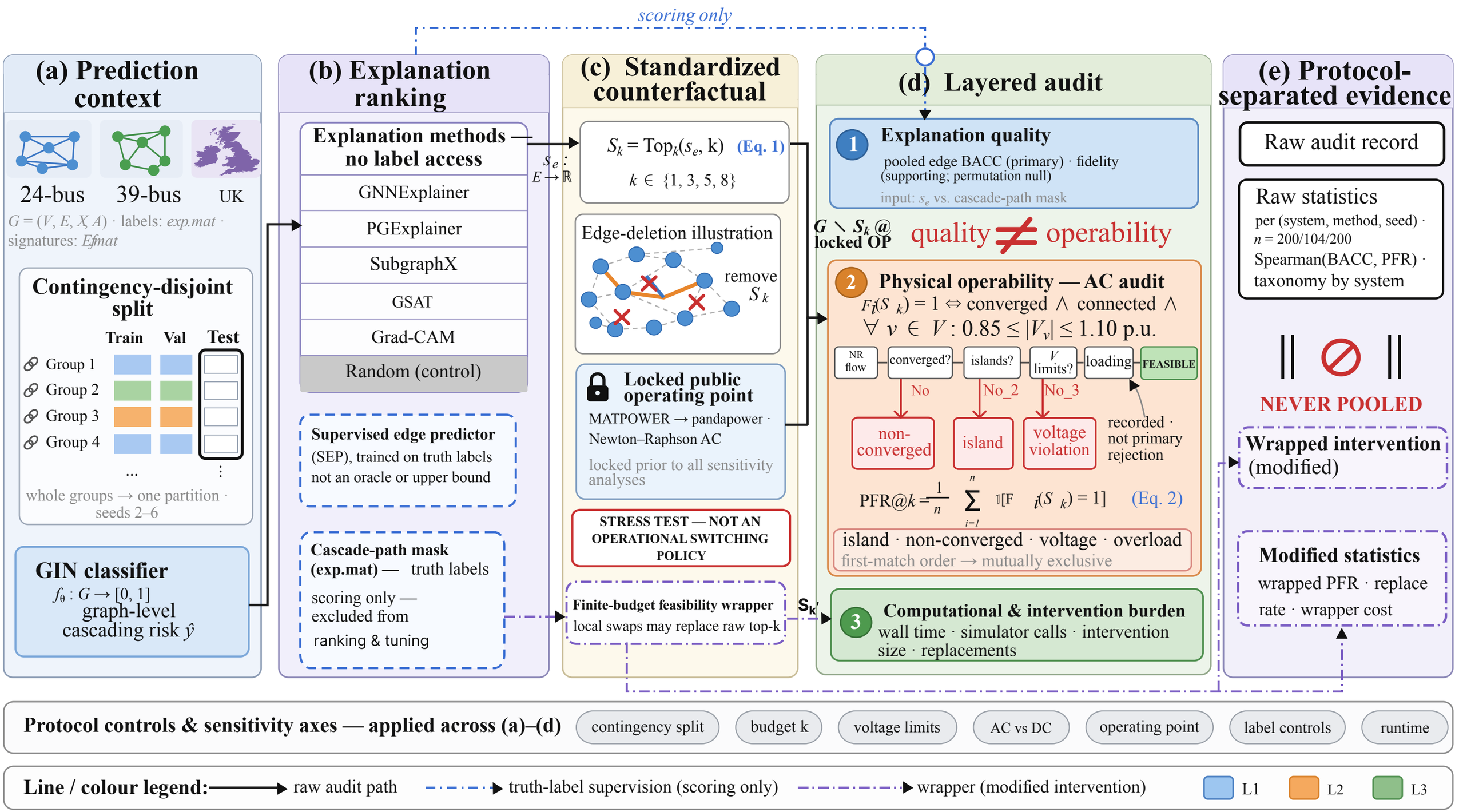

scoring only
(a) Prediction context
24-bus
39-bus
UK
G = (V, E, X, A) · labels: exp.mat · signatures: Efmat
Contingency-disjoint split
Train
Val
Test
Group 1
Group 2
Group 3
Group 4
whole groups → one partition · seeds 2–6
GIN classifier
f_θ : G → [0, 1]
graph-level cascading risk ŷ
(b) Explanation ranking
Explanation methods — no label access
GNNExplainer
PGExplainer
SubgraphX
GSAT
Grad-CAM
Random (control)
s_e : E → ℝ
Supervised edge predictor
(SEP), trained on truth labels
not an oracle or upper bound
Cascade-path mask (exp.mat) — truth labels
scoring only — excluded from ranking & tuning
(c) Standardized counterfactual
S_k = Top_k(s_e, k) (Eq. 1)
k ∈ {1, 3, 5, 8}
Edge-deletion illustration
remove S_k
Locked public operating point
MATPOWER → pandapower · Newton–Raphson AC
locked prior to all sensitivity analyses
STRESS TEST — NOT AN OPERATIONAL SWITCHING POLICY
Finite-budget feasibility wrapper
local swaps may replace raw top-k
S_k'
(d) Layered audit
1 Explanation quality
pooled edge BACC (primary) · fidelity (supporting; permutation null)
input: s_e vs. cascade-path mask
G \ S_k @ locked OP
quality ≠ operability
2 Physical operability — AC audit
F_i(S_k) = 1 ⇔ converged ∧ connected ∧ ∀ v ∈ V : 0.85 ≤ |V_v| ≤ 1.10 p.u.
NR flow
converged?
islands?
V limits?
loading
FEASIBLE
No
No_2
No_3
non-converged
island
voltage violation
recorded · not primary rejection
PFR@k = 1/n Σ_{i=1}^{n} 𝟙[F_i(S_k) = 1] (Eq. 2)
island · non-converged · voltage · overload
first-match order → mutually exclusive
3 Computational & intervention burden
wall time · simulator calls · intervention size · replacements
(e) Protocol-separated evidence
Raw audit record
Raw statistics
per (system, method, seed) · n = 200/104/200
Spearman(BACC, PFR) · taxonomy by system
NEVER POOLED
Wrapped intervention (modified)
Modified statistics
wrapped PFR · replace rate · wrapper cost
Protocol controls & sensitivity axes — applied across (a)–(d)
contingency split
budget k
voltage limits
AC vs DC
operating point
label controls
runtime
Line / colour legend:
raw audit path
truth-label supervision (scoring only)
wrapper (modified intervention)
L1
L2
L3